

CSC-696D, Takanori Fujiwara

Final project proposal. Novel visualization techniques for git data.

0.1 Problem description

In this project we will create several novel visual analytic systems for git data. The tools we develop will help developers quickly gain an understanding of git repositories. We will focus on a few key questions.

1. Who are the main contributors to a repository
2. At what times was development most active for any repository
3. Can we identify times when AI assisted development became popular on a given repository
4. What is the best visualization for giving developers an overview of the project structure

0.2 Problem description and motivation

Developers often work with large amounts of code and are forced to quickly learn the custom's and conventions of many repositories, this process can be very time-consuming. By giving developers tools for quickly visualizing the history of a repository we can immediately give them an understanding of conventions, key contributors and etiquette.

Many visualizations have been developed over the years to work with git data, in this project we want to develop a handful of new visualizations by first studying visualizations that others have developed and refining them to improve their visual clarity, augmenting them with additional data or even combining two visualization techniques into one.

0.3 Related research papers

- Githru: Visual Analytics for Understanding Software Development History Through Git Metadata Analysis
- Git-Truck: Hierarchy-Oriented Visualization of Git Repository Evolution
- ChangePrism: Visualizing the Essence of Code Changes

0.4 Planned technical approach

0.5 Selected dataset(s)

0.6 Expected results and evaluation methodology

- evaluation examples include: case studies, user studies, quantitative evaluation